

YILDIZ TECHNICAL UNIVERSITY
MATHEMATICAL ENGINEERING
NUMERICAL ANALYSIS FINAL EXAM
2020 SUMMER SCHOOL

31.08.2020

- 1) Show that $g(x) = \pi + 0.5 \sin\left(\frac{x}{2}\right)$ has a unique fixed point on $[0, 2\pi]$. Use fixed point iteration to find an approximation to the fixed point that is accurate to within 10^{-2}

- 2) Find the value of x for $y=3.524$ using divided differences table for the following data.

x	0.0	0.5		1.0	1.5	2.0
$f(x)$	1.000	1.648		2.718	4.481	7.386

- 3) Use the Romberg Integration method and the following data to approximate $\int_1^5 f(x)dx$. Notice: Please use all data to determine the dimension of R matrix.

x	1	2	3	4	5
$f(x)$	2.4142	2.6734	2.8974	3.0976	3.2804

- 4) Use the data linearization method (Least Squares Method) to fit the curve of the form $y = ab^x$ for the data given below.

x_k	1	2	3	4	5
y_k	130	150	175	190	240

Note: Use three digits after the decimal point.